



S.A.F.A.L

Where Smart Monitoring Meets Urban

Safety

Smart Aqua Flood Alert

Layer

| Reactive Chaos

The Urban Flood Crisis

Our cities are drowning in inefficiency. Traditional methods are passive, leading to costly disasters.

- ⚠️ **Delayed Response:** Action is taken only after damage occurs.
- 🚗 **Safety Hazards:** Traffic gridlock and compromised road safety.
- 💰 **Economic Loss:** Billions lost in property damage annually.



Introducing S.A.F.A.L

**A comprehensive Smart Aqua Flood Alert Layer
designed to transform how cities handle water.**



Smart
Layer



Alert
Layer



Safety
Layer

| Technology Core

Smart Monitoring

IoT Sensor Grid: A network of ultrasonic sensors placed in drains monitors water levels in real-time, 24/7.

Predictive AI: Analyzes historical data and weather forecasts to predict flood risks hours in advance.

Instant Alerts

Automated Triggers: When thresholds are breached, the system instantly triggers automated pumps.

Emergency Broadcast: Pushes immediate notifications to municipal dashboards and public apps.

Sustainable Infrastructure



Permeable Concrete

Porous pavement allows surface water to infiltrate directly into the ground, reducing runoff by up to 40%.



Green Roofs

Vegetated rooftops absorb rainfall at the source, delaying peak flow and improving urban air quality.

| How S.A.F.A.L Works



1. Detect

Heavy rain is forecast and sensors detect rising water levels in drains.



2. Process

AI analyzes the rate of rise vs. drainage capacity to predict overflow.



3. Act

Pumps activate automatically, and alerts are sent to citizens.

| Citizen Safety App

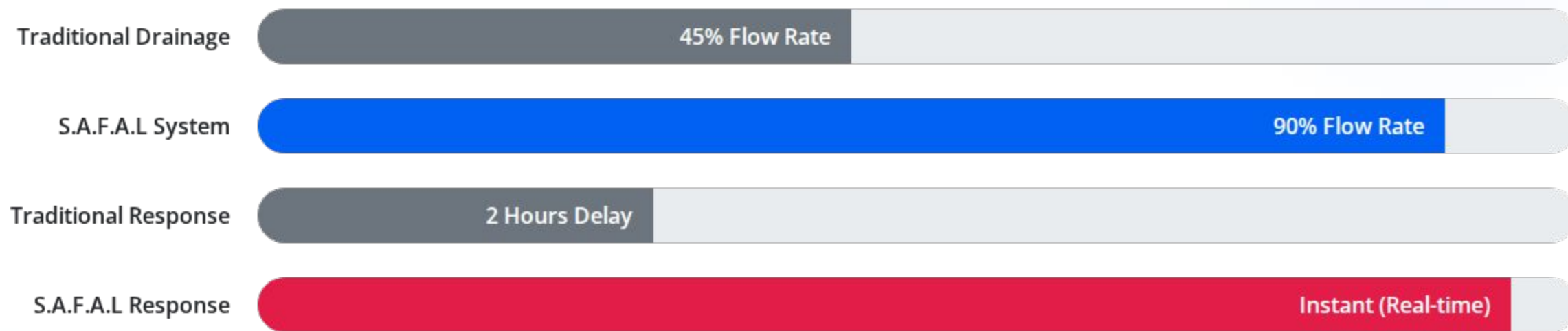


Real-Time Awareness

Empowering citizens with the information they need to stay safe.

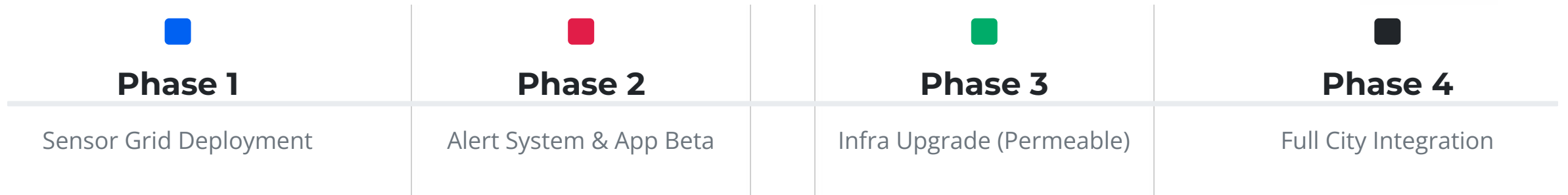
- **Live Maps:** Visualize waterlogged zones in real-time.
- **Route Diverts:** Get AI-suggested safe routes to avoid flooded roads.
- **Emergency SOS:** Direct line to emergency services with location sharing.

Efficiency Comparison



The integration of automated pumps and predictive AI doubles drainage efficiency.

| Implementation Plan



**"We are moving from reactive chaos to
proactive resilience."**

- S.A.F.A.L Mission

Questions?

Let's make our cities S.A.F.A.L together.

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| Image Sources



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Source: elevenmyanmar.com



<https://lifestyle.sustainability-directory.com/wp-content/uploads/2025/07/close-up-macro-shot-of-concrete-surface-with-granular-salt-and-water-droplets.jpg>

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